

AI/BigData 인텐시브 과정

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Section
코스소개

DAY1

머신러닝
기초 개념

DAY2

지도학습

DAY3

DAY4

서비스
개발

DAY5

앙상블 학습

DAY6

비지도학습

DAY7

시계열분석

DAY8

DAY9

프로젝트

DAY10

□ 기초 수학 기호

□ 미분

AI/BigData 인텐시브 과정

Section 1. 기초 수학

Section 1-1. 인공지능을 위한 기초 수학

수학 기호

$<$

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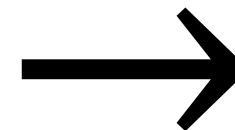
$\leq$

$\geq$

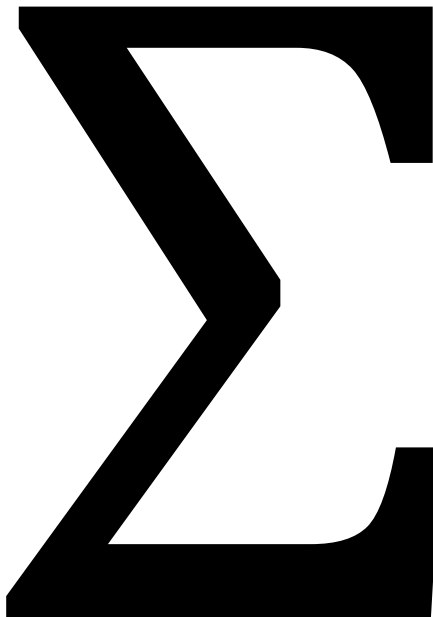
$-$

$>$

수학 기호

 Σ $\int$ $\min x$  ∂  Δ $\propto$ $\operatorname{argmin} x$

시그마(Sigma)



시그마(Sigma)

The diagram illustrates the summation formula $\sum_{x=1}^n x = 1 + 2 + \dots + n$. The summation symbol $\sum$ is the central element. Above it, the variable n is enclosed in an orange circle, with an upward-pointing arrow and the Korean word '끝' (end) above that. Below the summation symbol, the expression $x=1$ is shown, with x in a blue circle and $=1$ in a green circle; a downward-pointing arrow and the Korean word '시작' (start) are below this. To the right of the summation symbol, the variable x is in a blue circle, followed by an equals sign. The right side of the equation is $1 + 2 + \dots + n$, where the '1' and 'n' are in green and orange circles respectively, and the ellipsis consists of three dots.

$$\sum_{x=1}^n x = 1 + 2 + \dots + n$$

시그마(Sigma)

$$\sum_{x=1}^{10} x = 1 + 2 + \dots + 10$$

$$\sum_{x=7}^9 x = 7 + 8 + 9$$

$$\sum_{x=1}^{10} 3 = 3 + 3 + \dots + 3$$

```
[1]: res = 0
     for i in range(1, 11):
         res = res + i
```

```
[2]: print(res)
```

55

```
[3]: res = 0
     for i in range(7, 10):
         res = res + i
```

```
[4]: print(res)
```

24

```
[5]: res = 0
     for i in range(1, 11):
         res = res + 3
```

```
[6]: print(res)
```

30

파이(pi)

Π

파이(pi)

The diagram illustrates the factorial formula $x! = 1 \times 2 \times \dots \times n$ with visual annotations. The variable x is in a blue circle, and the result n is in an orange circle. The numbers 1 and 2 are in green circles. The product is shown as $x! = 1 \times 2 \times \dots \times n$. An arrow points from the word '끝' (end) to the orange n , and another arrow points from the word '시작' (start) to the green 1. The variable x is also annotated with a blue circle containing x and a green circle containing 1, with an equals sign between them.

$$\prod_{x=1}^n x = 1 \times 2 \times \dots \times n$$

끝

시작

파이(pi)

$$\prod_{x=1}^{10} x = 1 \times 2 \times \cdots \times 10$$

```
[9]: res = 1
    for i in range(1, 11):
        res = res * i
```

```
[10]: print(res)
```

3628800

```
[11]: 1*2*3*4*5*6*7*8*9*10
```

```
[11]: 3628800
```

파이(pi)

1.9. Naive Bayes

Naive Bayes methods are a set of supervised learning algorithms based on applying Bayes' theorem with the "naive" assumption of conditional independence between every pair of features given the value of the class variable. Bayes' theorem states the following relationship, given class variable y and dependent feature vector x_1 through x_n :

$$P(y | x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n | y)}{P(x_1, \dots, x_n)}$$

Using the naive conditional independence assumption that

$$P(x_i | y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = P(x_i | y),$$

for all i , this relationship is simplified to

$$P(y | x_1, \dots, x_n) = \frac{P(y) \prod_{i=1}^n P(x_i | y)}{P(x_1, \dots, x_n)}$$

Since $P(x_1, \dots, x_n)$ is constant given the input, we can use the following classification rule:

$$P(y | x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i | y)$$

$$\Downarrow$$

$$\hat{y} = \arg \max_y P(y) \prod_{i=1}^n P(x_i | y),$$

지시 함수(indicator function)

$$I_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

x 가 A 에 속하면 1

x 가 A 에 속하지 않으면 0

$$1\{y_i = \text{사과}\} = \begin{cases} 1, & y_i = \text{사과} \\ 0, & y_i \neq \text{사과} \end{cases}$$

y_i 가 사과면 1

y_i 가 사과가 아니면 0

지시 함수(indicator function)

$$I_{x < 10}(13) = 0$$

```
[25]: x = 13
      if x < 10:
          res = 1
      else:
          res = 0
```

```
[26]: print(res)
```

0

지시 함수(indicator function)

피쳐(feature)

타겟(target)

둘레 x1	당도 x2	과일 종류 y
30	40	수박
20	20	참외
10	17	사과
15	15	사과

$$1\{y_3 = \text{사과}\} = 1 \quad \text{3번째 타겟 값이 사과인지 여부}$$

$$\sum_{i=1}^4 1\{y_i = \text{사과}\} = 2 \quad \text{사과 개수}$$

$$1\{y_1 = \text{사과}\} + 1\{y_2 = \text{사과}\} + 1\{y_3 = \text{사과}\} + 1\{y_4 = \text{사과}\} \\ = 0 + 0 + 1 + 1 = 2$$

$$\frac{1}{4} \sum_{i=1}^4 1\{y_i = \text{사과}\} = 0.5 \quad \text{사과일 확률}$$

지시 함수(indicator function)

<https://docs.pytorch.org/docs/stable/generated/torch.nn.CrossEntropyLoss.html>

PyTorch v2.11.... Install PyTorch User Guide Reference API Developer Notes Community Tutorials Go to pytorch.org Search the docs ...

EmbeddingBag CosineSimilarity PairwiseDistance L1Loss MSELoss **CrossEntropyLoss**

Home > Reference API > torch.nn > CrossEntropyLoss Rate this Page ★★★★★

CrossEntropyLoss

```
class torch.nn.CrossEntropyLoss(weight=None, size_average=None, ignore_index=-100, reduce=None, reduction='mean',
                                label_smoothing=0.0)
```

[\[source\]](#)

$$\ell(x, y) = L = \{l_1, \dots, l_N\}^\top, \quad l_n = -w_{y_n} \log \frac{\exp(x_{n,y_n})}{\sum_{c=1}^C \exp(x_{n,c})} \cdot 1\{y_n \neq \text{ignore_index}\}$$

where x is the input, y is the target, w is the weight, C is the number of classes, and N spans the minibatch dimension as well as $d_1, \dots, d_k$ for the K -dimensional case. If **reduction** is not **'none'** (default **'mean'**), then

$$\ell(x, y) = \begin{cases} \sum_{n=1}^N \frac{1}{\sum_{n=1}^N w_{y_n} \cdot 1\{y_n \neq \text{ignore_index}\}} l_n, & \text{if reduction = 'mean';} \\ \sum_{n=1}^N l_n, & \text{if reduction = 'sum'}. \end{cases}$$

자연 상수 e

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

$$e = 2.7182818284$$

1.9.1. Gaussian Naive Bayes

`GaussianNB` implements the Gaussian Naive Bayes algorithm for classification. The likelihood of the features is assumed to be Gaussian:

$$P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

The parameters σ_y and μ_y are estimated using maximum likelihood.

```
>>> from sklearn.datasets import load_iris
>>> from sklearn.model_selection import train_test_split
>>> from sklearn.naive_bayes import GaussianNB
>>> X, y = load_iris(return_X_y=True)
>>> X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.5, random_state=0)
>>> gnb = GaussianNB()
>>> y_pred = gnb.fit(X_train, y_train).predict(X_test)
>>> print("Number of mislabeled points out of a total %d points : %d"
...       % (X_test.shape[0], (y_test != y_pred).sum()))
Number of mislabeled points out of a total 75 points : 4
```

min, argmin

$\min x$

$\operatorname{argmin} x$

min, argmin

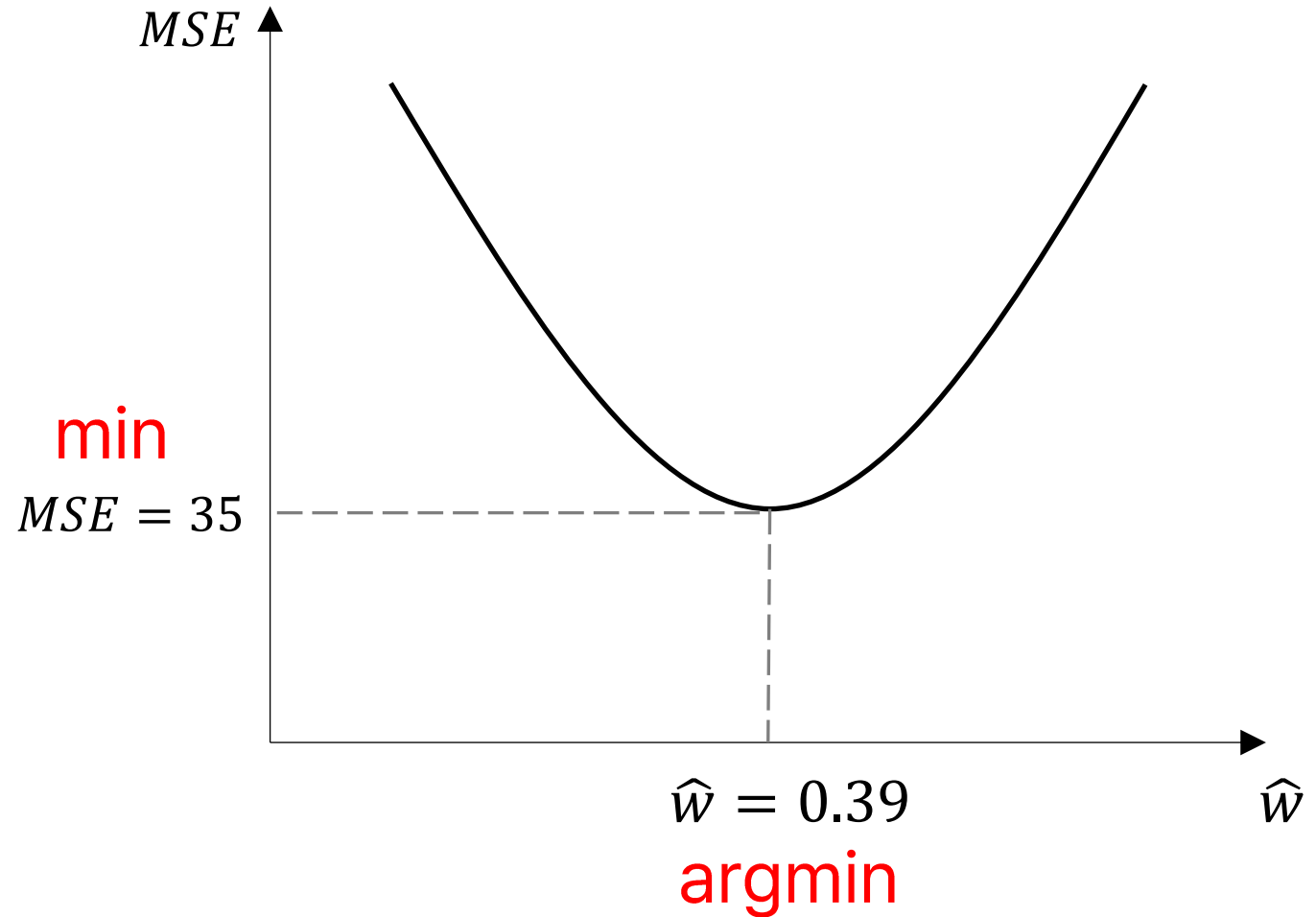
$$\begin{array}{cccccc} \text{인덱스} & 0 & 1 & 2 & 3 & 4 \\ x & = & [4, 2, 5, 9, 3] \end{array}$$

$$\min x = 2$$

$$\operatorname{argmin} x = 1$$

min, argmin

⋮	⋮
$\hat{w} = 0.39$	$MSE = 35$
$\hat{w} = 0.40$	$MSE = 86$
$\hat{w} = 0.41$	$MSE = 127$
⋮	⋮



min, argmin

$$\underset{\hat{w}}{\operatorname{argmin}}\{\text{목적함수}\}$$

min, argmin

$$\operatorname{argmin}_{\hat{w}} \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$$

$$= \operatorname{argmin}_{\hat{w}} \frac{1}{n} \sum_{i=1}^n (y - (\hat{w}x + b))^2$$

'머신러닝을 배운다'의 의미

$$\operatorname{argmin}_{\hat{w}} \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$$
$$\operatorname{argmin}_{\hat{w}} \frac{1}{n} \sum_{i=1}^n (y - \text{알고리즘1})^2$$
$$\operatorname{argmin}_{\hat{w}} \frac{1}{n} \sum_{i=1}^n (y - \text{알고리즘2})^2$$
$$\vdots$$
$$\operatorname{argmin}_{\hat{w}} \frac{1}{n} \sum_{i=1}^n (y - \text{알고리즘10})^2$$

min, argmin

인덱스 0 1 2 3 4
 $x = [4, 2, 5, 9, 3]$



최소값

```
[33]: x = [4, 2, 8, 9, 3]
```

```
[34]: min_value = min(x)
```

```
[35]: print(min_value)
```

2

```
[36]: argmin_value = x.index(min_value)  
print(argmin_value)
```

1

```
[37]: import numpy as np  
min_value = np.min(x)  
print(min_value)
```

2

```
[38]: argmin_value = np.argmin(x)  
print(argmin_value)
```

1

max, argmax

인덱스 0 1 2 3 4
 $x = [4, 2, 5, 9, 3]$
 ↑
 최대값

```
[39]: x = [4, 2, 8, 9, 3]
      max_value = max(x)
      print(max_value)
```

9

```
[40]: argmax_value = x.index(max_value)
      print(argmax_value)
```

3

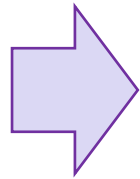
```
[41]: import numpy as np
      max_value = np.max(x)
      print(max_value)
```

9

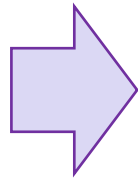
```
[43]: argmax_value = np.argmax(x)
      print(argmax_value)
```

3

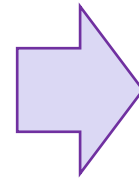
max, argmax



AI
Model



독수리	0.95
표범	0.03
상어	0.02

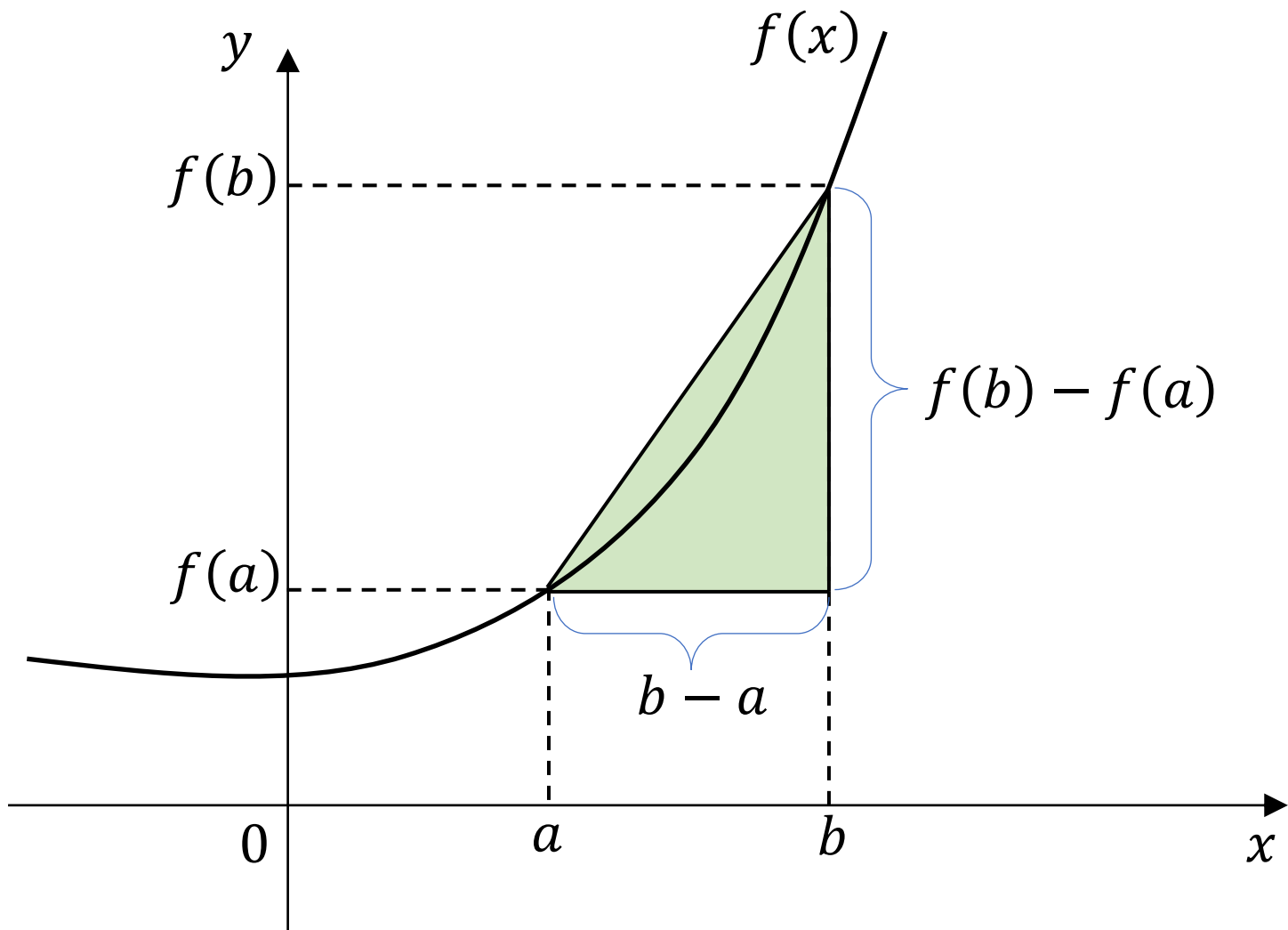


독수리

$$\max x = 0.95$$

$$\operatorname{argmax} x = \text{독수리}$$

미분(differential)

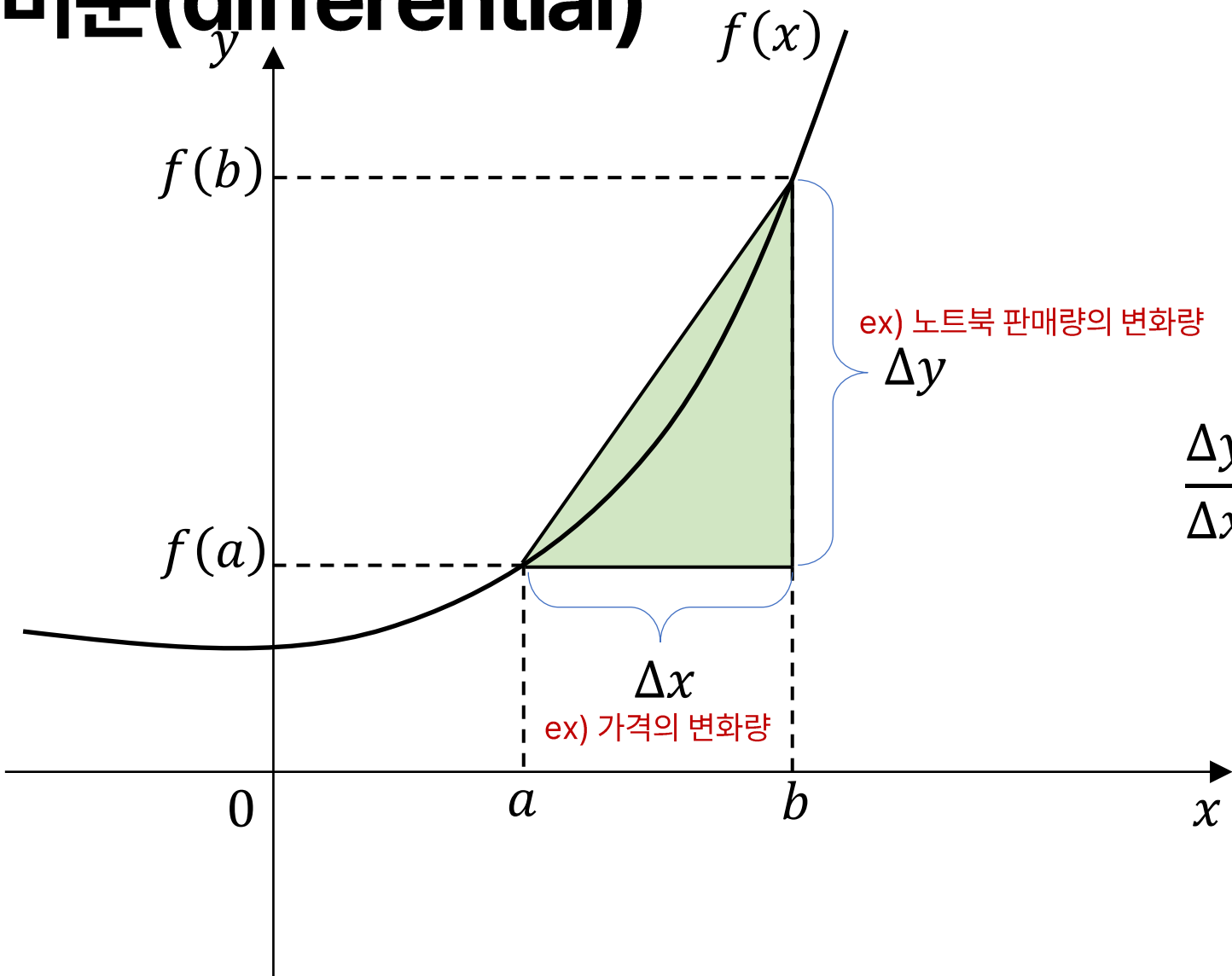


x 값이 a 에서 b 로 변할때
 y 값이 $f(a)$ 에서 $f(b)$ 로 변함

평균 변화율

$$\frac{f(b) - f(a)}{b - a}$$

미분(differential)

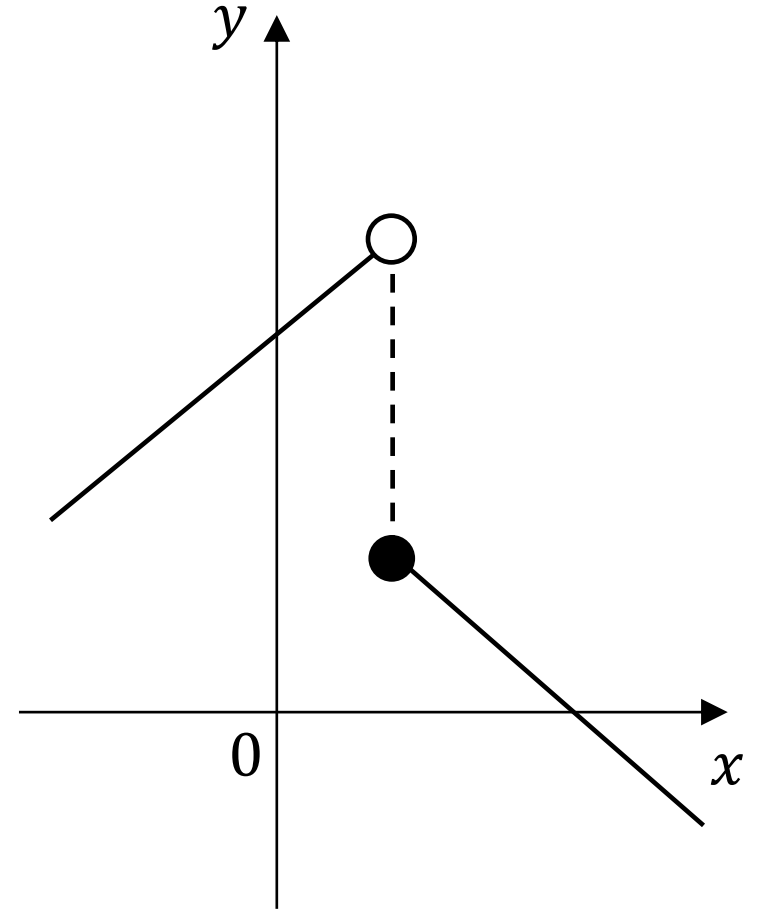
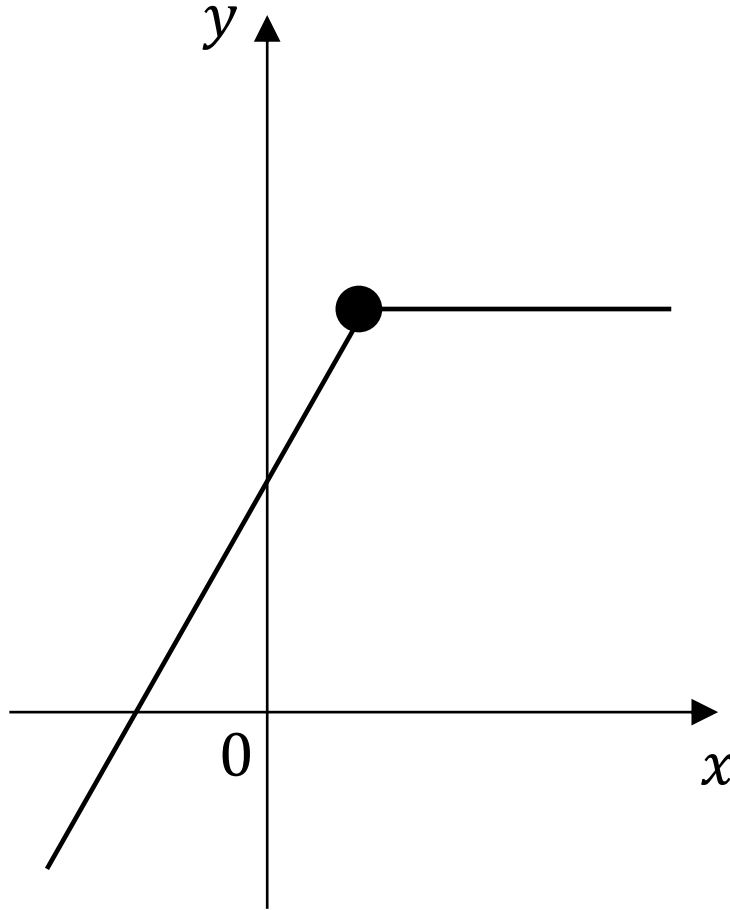
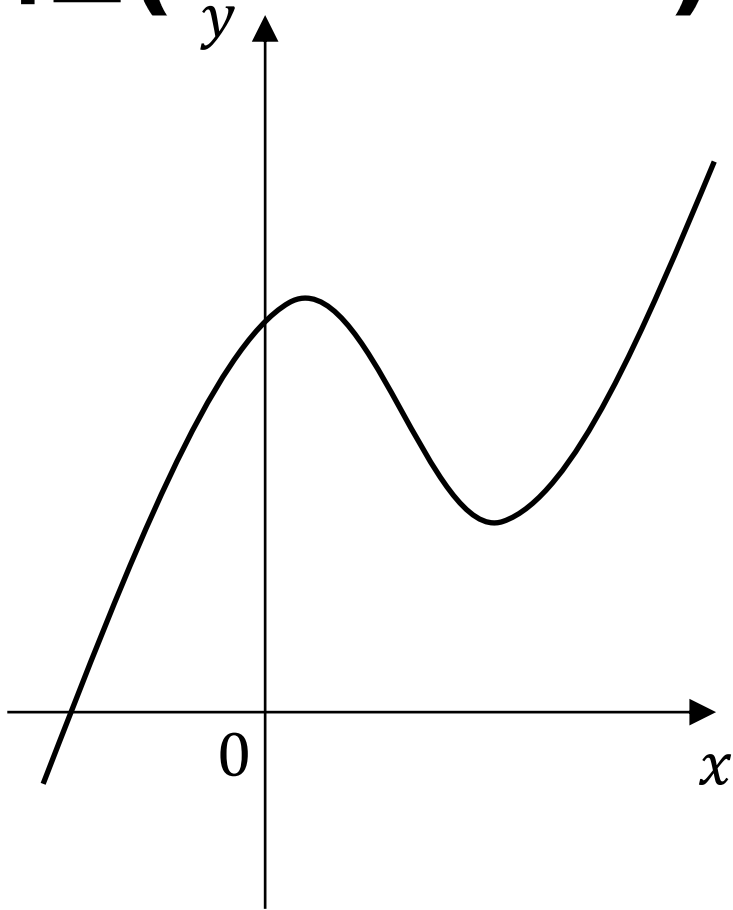


$$\Delta x = b - a$$

$$\Delta y = f(b) - f(a)$$

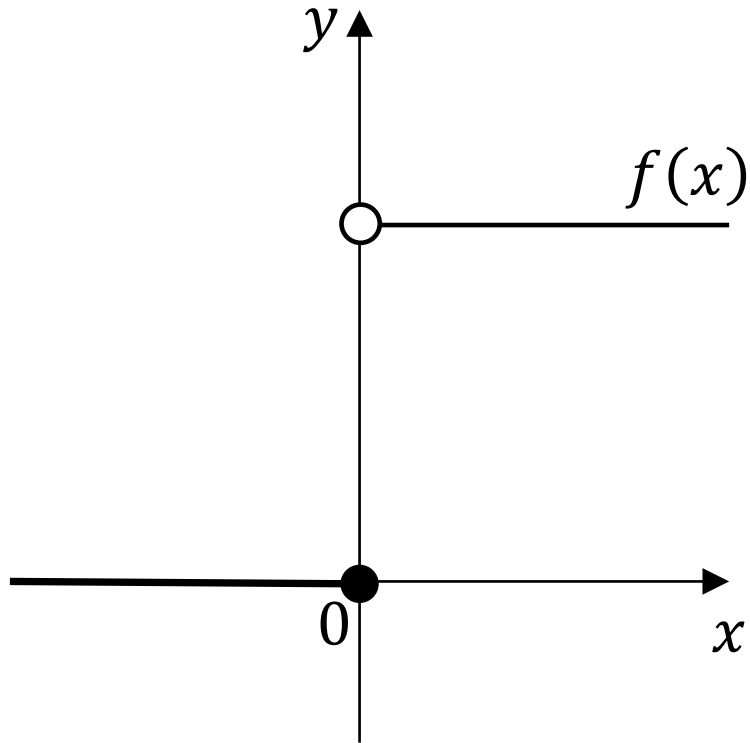
$$\frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a} = \frac{f(a + \Delta x) - f(a)}{\Delta x}$$

미분(differential)

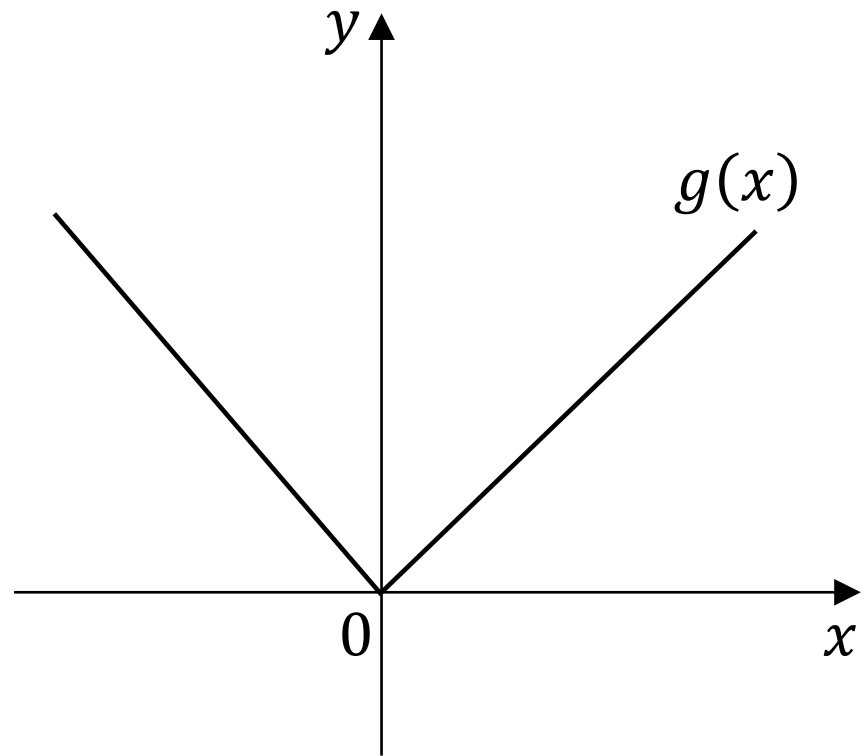


모든 점에서 미분 가능하려면 뾰족하지 않고 끊어지면 안된다

미분 불가능

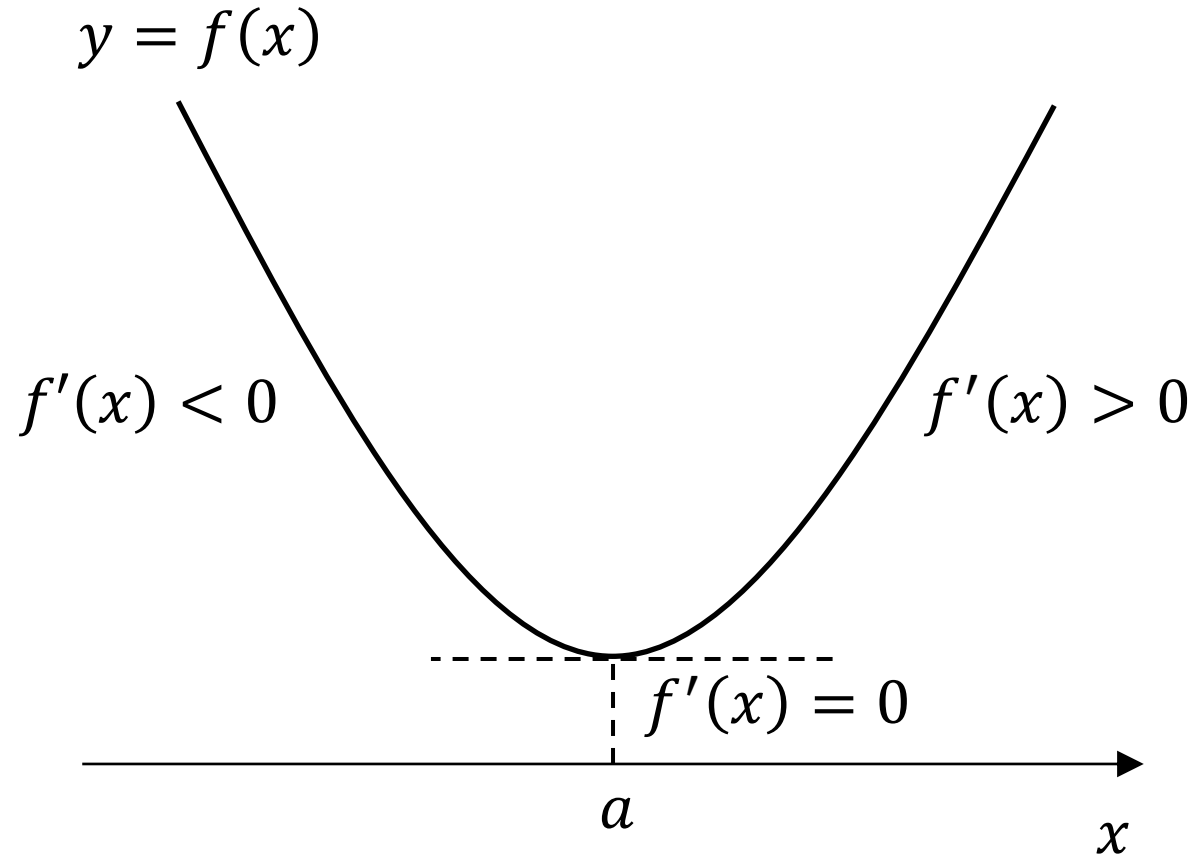


$$f(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$$



$$g(x) = |x|$$

미분(differential)



편미분(partial derivative)

다변수 함수에서 하나의 변수에 대한 변화율을 측정하는 방법 → 특정 방향으로의 기울기를 의미

↑ ↑ ↑ ↑ 핸드폰 판매량 가격 성능 가격 함수 $y = f(x_1, x_2)$가 존재할때, x_1에 대한 편미분	함수 $f(x_1, x_2)$가 존재할때, x_2에 대한 편미분
↑ ↑ 100만원 10만원 $\frac{\partial f}{\partial x_1} = \lim_{\Delta x_1 \rightarrow 0} \frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1}$	$\frac{\partial f}{\partial x_2} = \lim_{\Delta x_2 \rightarrow 0} \frac{f(x_1, x_2 + \Delta x_2) - f(x_1, x_2)}{\Delta x_2}$
x2를 고정하고 x1를 조금 변화시켰을 때 함수 f의 변화율	x1를 고정하고 x2를 조금 변화시켰을 때 함수 f의 변화율

편미분(partial derivative) - 예제

함수 $f(x_1, x_2) = x_1^2 + x_2^2$

x_1 에 대한 편미분

$$\frac{\partial f}{\partial x_1} = \frac{\partial}{\partial x_1} (x_1^2 + x_2^2) = 2x_1$$

x_2 에 대한 편미분

$$\frac{\partial f}{\partial x_2} = \frac{\partial}{\partial x_2} (x_1^2 + x_2^2) = 2x_2$$

그레디언트(gradient)

편미분이 다변수 함수에서 하나의 변수에 대한 변화율을 측정하는 것이라면,
그레디언트는 다변수 함수의 모든 편미분을 벡터 형태로 묶어 표현한 것

→ 각 변수에 대한 편미분을 한 곳에 모은 벡터

함수 $f(x_1, x_2)$ 의 그레디언트

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right)$$

그레디언트(gradient)

함수 $f(x_1, x_2, \dots, x_n)$ 의 그레디언트

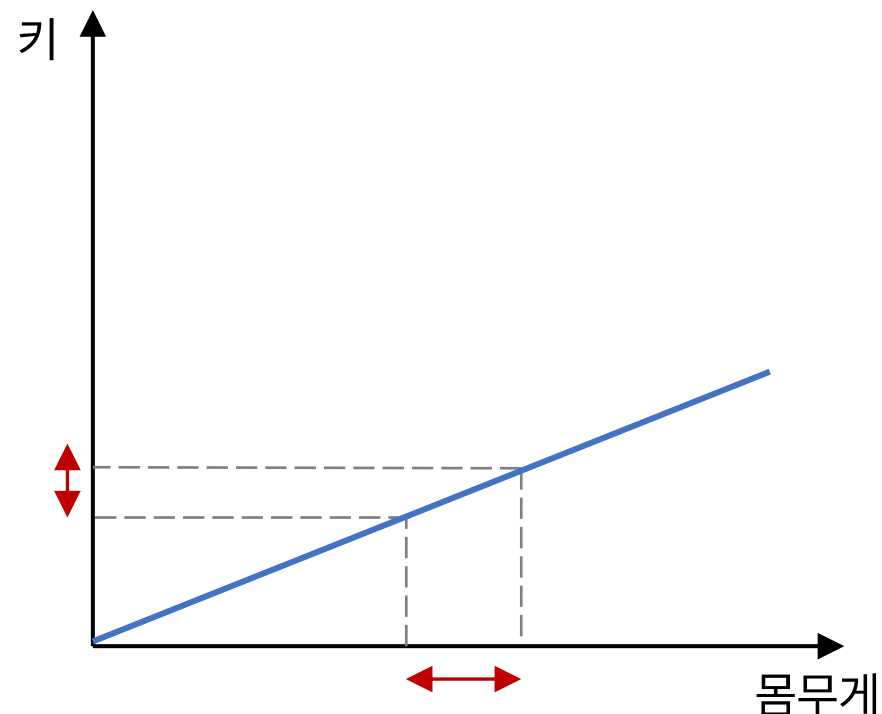
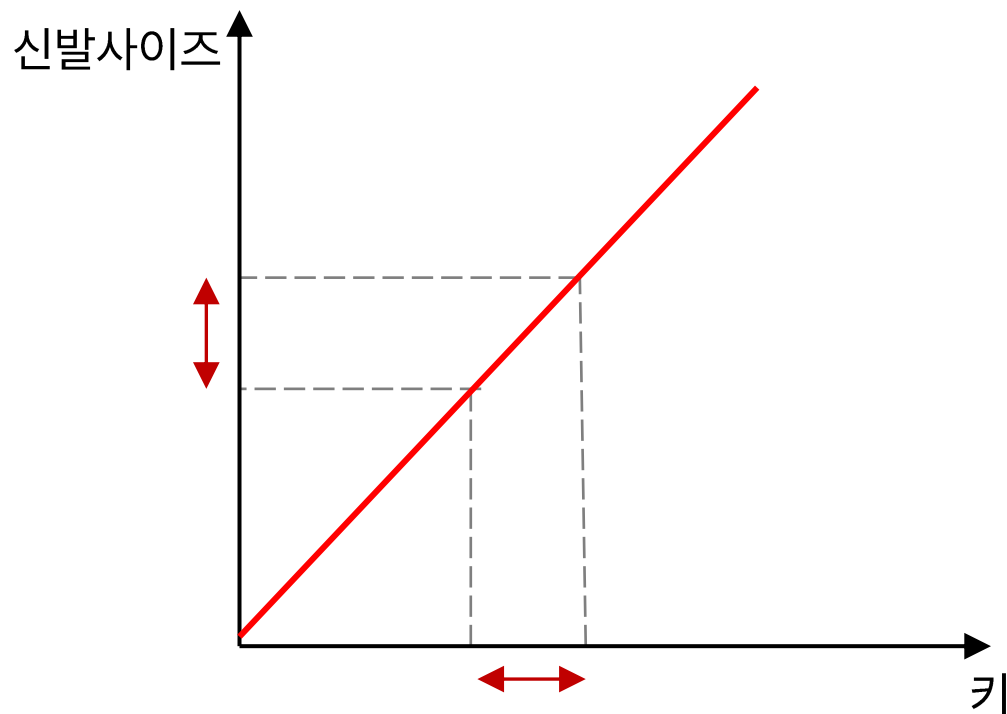
$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)$$

연쇄법칙(chain rule)

$$\frac{dy}{dx_1} = \frac{dy}{dx_2} \cdot \frac{dx_2}{dx_1}$$

연쇄법칙(chain rule)

$$\frac{d\text{신발사이즈}}{d\text{몸무게}} = \frac{d\text{신발사이즈}}{d\text{키}} \cdot \frac{d\text{키}}{d\text{몸무게}}$$



적분(integral)

$$F'(x) = f(x)$$

$$F(x) = \int f(x) dx$$

감사합니다.

Q & A